AI-Powered Behavioral Anomaly Detection for Cybersecurity

Vajra AI — Cascaded Behavioral Threat Detection & Response

Attribute Details
Problem Statement Title AI-Powered Behavioral Anomaly Detection for Cybersecurity
Theme Cybersecurity / Smart Automation (TO BE CONFIRMED BY USER)
PS Category Software
Problem Statement ID TO BE FILLED — USER MUST PROVIDE (Check Hackathon Portal)
Student Name TO BE FILLED — USER MUST PROVIDE
Student ID TO BE FILLED — USER MUST PROVIDE

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Idea Title: Proposed Solution

Vajra AI — Cascaded Behavioral Threat Detection & Response

System Overview
• Models 14-day trailing access baselines per entity without hardcoded IP blacklists.
• Detects complex multi-stage attacks via a 2-tier Staged Cascade Pipeline.
• Maps anomalies directly to official MITRE ATT&CK; techniques with remediation steps.
• Displays operational Risk (0-100) vs. ML Confidence (0-100%) on a Streamlit SOC Console.

Addressing Problem Statement Requirements
• Sequential Behavior: Tracks K=5 event windows and 9D deviation feature matrix.
• Extreme Class Imbalance: Evaluated with balanced Random Forest and alert budget capacity.
• Concept Drift: Updates baselines via Exponential Moving Average (alpha=0.05) on benign logs.
• Explainability & Cold-Start: SHAP feature drivers + peer-group fallback (70% confidence cap).

Innovation & Uniqueness
• Staged Cascade: Fast Isolation Forest cuts deep PyTorch LSTM workload by 96.88%.
• Physics-Rule Hybrid: Hard-coded velocity check (>900 km/h) flags impossible_travel with 100% accuracy.
• Alert Budget Tuning: Evaluates recall against analyst capacity (Top 1% = 75.61% Recall / 0.89% FPR).

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Technical Approach & Architecture

Technology Stack, Cascade Pipeline & 9D Feature Engine

Tech Stack Pipeline Flow Diagram

• Language: Python 3.12
• Data: pandas 2.1.4, NumPy 1.26.2
• ML Engine: scikit-learn 1.7.2
• Deep Learning: PyTorch 2.13.0
• Dashboard: Streamlit 1.60.0, Plotly
• Explainability: SHAP 0.44.1

[Log Stream] (91,805 Events)
■■■ [Baseline Profiler] (14-Day EMA, alpha=0.05) ■■■ [9D Feature Matrix]
■■■ [Stage 1: Isolation Forest] (Inliers ~97% Safe) ■■■ [Stage 2: PyTorch LSTM] (Candidates ~3%) ■■■ [Classifier & Physics Override]
■■■ [Risk & MITRE Engine] ■■■ [Streamlit UI]

Core 9D Baseline Deviation Features

1. geo_distance_km | 2. time_of_day_zscore | 3. resource_novelty | 4. session_duration_zscore | 5. auth_failure_rate_trailing

6. source_ip_entity_fanout | 7. command_sequence_novelty | 8. fingerprint_mismatch | 9. is_cold_start

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Feasibility & Viability Analysis

Benchmarked Performance, Technical Limitations & Mitigation Strategies

Benchmarked Working Prototype Results

• Prototype Load Latency: 4-page Streamlit SOC console loads complete pipeline data in 6.8s.

• Workload Reduction: Stage 1 filters 96.88% of logs; Stage 2 LSTM evaluates only 3.12% suspicious pool.

• Top 1.0% Alert Budget Performance (Temporal Split: Days 1-21 Train, Days 22-30 Test):

System Recall: 75.61% (31/41 attacks) | False Positive Rate: 0.89% (245 FPs) | Precision: 11.23%

Identified Technical Risks / Limitations Engineering Mitigation Strategies

• Base-Rate Fallacy: 0.15% attack prevalence causes false alarms to
outnumber true attacks.
• Asset Severity Bias: 10 low-asset attacks (sev=0.3) fall below top 1% risk
cutoff.
• Low Sample Sizes: 5 of 6 attack classes have N <= 5 in strict temporal
split.
• Untested Physics Range: Speed rule untested in 900-5000 km/h flight layover range.

• Alert Budget Tuning: Tune operating cutoffs to analyst review capacity rather than static rules.
• Dual Ranking Modes: Provide toggle between Severity-Risk Rank and Pure Anomaly Rank.
• Active Learning Loop: Log analyst feedback (Confirm/Dismiss) to retrain
baseline profiles.
• Real-World Calibration: Calibrate speed threshold against airline flight data
prior to deployment.

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Artifacts: Dashboard & Core Code Logic

Streamlit Console Views & Representative Pipeline Engine Code

Interactive Dashboard Screenshots Representative Engine Code Snippet

1. `reports/screenshots/01_home_dashboard.png`

• Header, LIVE indicator, pipeline status strip, KPIs.

2. `reports/screenshots/02_live_threat_feed.png`

• Ranked alert queue table, inline reasons, ■ LIVE mode.

3. `reports/screenshots/03_triage_inspector.png`

• SHAP attributions, Risk/Confidence gauges, MITRE checklist.

4. `reports/screenshots/04_threat_analytics.png`

• Cascade benchmark charts & alert budget trade-off table.

File: utils/data_loader.py (Physics Override & Risk Engine)

# 1. Physics Impossible-Travel Hard Override Check is_physics_impossible = (feat_dict.get('is_physics_impossible_travel', 0.0) == 1.0)
if is_physics_impossible:
pred_attack = 'impossible_travel'
conf_score = 99.9
sev_label = 'CRITICAL'
risk_score = max(risk_score, 88.0)

2. Composite Risk Score Formula (0-100)
Risk = 40% Anomaly + 35% Max Deviation + 25% Severity Weight risk_score = (0.40 * norm_anom) + (0.35 * max_dev) + (0.25 * sev_weight *
100.0)
risk_score = float(np.clip(risk_score, 0.0, 100.0))

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Research & References

Standards, Cited Literature & Architectural Justifications

Domain / Method Reference & Architectural Justification
MITRE ATT&CK; Framework Standard taxonomy for adversary tactics and techniques (https://attack.mitre.org/). Used to map detected threats to
Technique IDs (T1110, T1036, T1078, T1021, T1041).
Stage 1: Isolation Forest Liu, F. T., Ting, K. M., & Zhou, Z. H. (2008). Isolation Forest. IEEE ICDM. Justifies low-latency ($<1.0\text{ms}$)
tabular filtering on 100% of incoming logs.
Stage 2: PyTorch LSTM Malhotra, P., et al. (2016). LSTM-based Encoder-Decoder for Multi-sensor Anomaly Detection. arXiv:1607.00148.
Justifies LSTM sequence autoencoders over Transformers for short K=5 windows.
EMA Concept Drift Gama, J., et al. (2014). A survey on concept drift adaptation. ACM Computing Surveys. Justifies updating trailing
baseline profiles with alpha=0.05 on confirmed benign traffic.
Explainability (SHAP) Lundberg, S. M., & Lee, S. I. (2017). A Unified Approach to Interpreting Model Predictions. NIPS. Justifies
game-theoretic feature attribution scoring for SOC triage cards.

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